

Total No. of Printed Pages: 03

B. E. - (Computer) (Sem-VII) (Revised Course 2019-2020)**EXAMINATION MAY 2024****CE710 Compiler Design****[Time: 3:00 Hours]****[Max. Marks: 100]****Instructions:**

- 1) Answer any 5 questions by selecting 2 questions from Part-A and 2 questions from Part-B and 1 question from Part-C
- 2) Make suitable assumption only if required

Part-A

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|----|---|-----------|
| Q1 | (a) Show the output of each phase of the compiler for the following:
position = initial + rate * 60 | 10 |
| | (b) Explain input buffering | 04 |
| | (c) Explain compiler construction tools | 06 |
| Q2 | (a) What is left-factoring? How is left-factoring eliminated? Left-factor the following grammar:
S → aSSbS
S → aSaSb
S → abb
S → b | 06 |
| | (b) Define context free grammar for the following statements.
(a) for loop with at least one if statement inside for loop.
(b) if statement with one assignment statement inside if | 06 |
| | (c) Check if the following grammar is LL(1):
S → iEtS iEtSeS a
E → b | 08 |
| Q3 | (a) Explain the role of lexical analyser. | 05 |
| | (b) Write a lex program to validate keywords and identifiers. | 05 |
| | (c) Write a short note on YACC | 05 |
| | (d) Explain the computation of first and follow of a grammar | 05 |

Part-B

- Q4 (a) Explain syntax directed translation scheme for assignment statement **04**
 (b) Construct the canonical collection of LR(0) items and SLR(1) parsing table for the following grammar: **06**
 $S \rightarrow L = R \mid R$
 $L \rightarrow *R \mid id$
 $R \rightarrow L$
 (c) Construct operator precedence parsing table for the following grammar and parse the string: id+id **10**
 $E \rightarrow E + T \mid T$
 $T \rightarrow T * F \mid F$
 $F \rightarrow (E) \mid id$
- Q5 (a) Construct CLR(1) parsing table for the following grammar: **08**
 $S \rightarrow CC$
 $C \rightarrow cC \mid d$
 (b) Write the algorithm for partitioning three-address instructions into basic blocks. **06**
 (c) What are the applications of dags? Construct a dag for: **06**
 $a = b + c$
 $b = b - d$
 $c = c + d$
 $e = b + c$
- Q6 (a) What is peephole optimization? Discuss any three peephole optimization techniques with the help of examples. **06**
 (b) Explain the data structures used to implement symbol table **06**
 (c) Construct LALR parsing table for the following grammar: **08**
 $A \rightarrow (A)$
 $A \rightarrow a$

Part-C

- Q7 (a) Explain the applications of compiler technology **05**
 (b) Write the features of LEX **05**
 (c) What is left recursion? Explain the steps to eliminate left recursion **05**
 (d) Explain the role of parser with the help of a diagram in compiler construction. **05**

BE1331

- Q8
- | | |
|---|-----------|
| (a) Explain the recovery techniques of syntax analysis phase errors | 05 |
| (b) Explain quadruples and triples with an example | 05 |
| (c) Explain the DAG representation of basic blocks | 05 |
| (d) What is symbol table? Describe the contents of symbol table. | 05 |